WeIntern Pvt Ltd

Data Science

Week 2 Internship Assignment Handbook

Domain: Data Science | Total Tasks: **3** | Deadline: **End of Week 2**

Task Overview

#	Task Name	Description
1	**Data Cleaning Project**	Clean and preprocess a raw dataset handling nulls, duplicates, and formatting issues.
2	**Exploratory Data Analysis**	Statistical analysis and visualizations including histograms, correlation matrices, and trends.
3	**Sales Prediction Model**	Build and evaluate a machine learning regression model to predict future sales.

Detailed Task Descriptions

TASK 1

Data Cleaning Project

Objective

Take a raw, messy real-world dataset and transform it into a clean, analysis-ready format. This task tests your ability to identify and resolve common data quality issues using Python's Pandas library.

Key Requirements

- Load a raw dataset (CSV or Excel) using Pandas
- Identify and handle missing/null values (drop, fill, or interpolate)
- Detect and remove duplicate rows
- Standardize column names and data types
- Handle outliers with IQR or Z-score methods
- Normalize or encode categorical variables where applicable
- Document every cleaning step with explanations

Implementation Steps

1. Load the dataset and perform initial exploration with .info(), .describe(), .head()
2. Visualize missing data using a heatmap (seaborn or missingno)
3. Decide and apply a strategy for each column with missing values
4. Identify and drop duplicate rows using .duplicated() and .drop_duplicates()
5. Rename columns to snake_case and correct data types
6. Detect outliers using box plots and IQR method
7. Encode categorical columns using label or one-hot encoding
8. Validate the cleaned dataset with summary statistics
9. Export the cleaned dataset to a new CSV file
10. Write a cleaning summary report

Recommended Tech Stack

Language	Python 3.x
Libraries	Pandas, NumPy, Seaborn, Matplotlib
Environment	Jupyter Notebook or Google Colab
Dataset Source	Kaggle, UCI ML Repository, or provided dataset

Deliverables

- Jupyter Notebook (.ipynb) with all steps documented
- Original raw dataset and cleaned output CSV
- Data cleaning report explaining decisions made
- Before and after comparison table of data quality metrics

Evaluation Criteria

- Completeness of cleaning steps
- Appropriateness of strategies for nulls, duplicates, and outliers
- Code readability and documentation quality
- Accuracy of data type corrections
- Quality of the final cleaned dataset

TASK 2

Exploratory Data Analysis

Objective

Perform a comprehensive Exploratory Data Analysis on a dataset to uncover patterns, trends, correlations, and insights. The goal is to summarize the data's main characteristics using both visual and statistical techniques.

Key Requirements

- Compute and interpret descriptive statistics for all columns
- Create at least 5 distinct types of visualizations
- Generate and analyze a correlation matrix
- Identify at least 3 actionable insights from the data
- Analyze distributions using histograms and KDE plots
- Perform grouped analysis by category or time period
- Present findings in a structured report format within the notebook

Implementation Steps

1. Load and inspect the cleaned dataset
2. Compute descriptive statistics (mean, median, std, quartiles)
3. Create histograms and KDE plots for numeric columns
4. Build a correlation heatmap for numeric features
5. Create box plots to visualize distribution and outliers by category
6. Analyze trends over time using line charts if time-series data exists
7. Perform group-by analysis on categorical variables
8. Create scatter plots to identify relationships between variables
9. Summarize findings with at least 3 data-driven insights
10. Write a conclusions section with business-relevant recommendations

Recommended Tech Stack

Language	Python 3.x
Libraries	Pandas, NumPy, Matplotlib, Seaborn, Plotly (optional)
Environment	Jupyter Notebook or Google Colab

Deliverables

- Jupyter Notebook with all analyses and visualizations
- At least 5 well-labeled charts with descriptions
- A written insights section within the notebook
- PDF export of the completed notebook

Evaluation Criteria

- Variety and appropriateness of visualizations
- Depth and accuracy of statistical analysis
- Quality and clarity of insights derived
- Notebook organization and narrative flow
- Chart labeling and presentation quality

TASK 3

Sales Prediction Model

Objective

Build a machine learning model that can accurately predict sales figures based on historical data. This task covers the full ML pipeline from feature engineering to model evaluation.

Key Requirements

- Load a sales dataset with multiple features
- Perform feature selection and engineering
- Split data into training and test sets (80/20)
- Train at least two regression models and compare performance
- Evaluate using RMSE, MAE, and R-squared metrics
- Visualize actual vs. predicted values
- Optimize the best model using hyperparameter tuning

Implementation Steps

1. Load and explore the sales dataset
2. Engineer relevant features (date components, lag features, rolling averages)
3. Handle missing values and encode categorical variables
4. Split data: 80% training, 20% testing
5. Train a Linear Regression model as a baseline
6. Train a Random Forest or Gradient Boosting model
7. Evaluate both models on the test set with RMSE, MAE, and R2
8. Visualize feature importance for the tree-based model
9. Plot actual vs. predicted values for the best model
10. Tune hyperparameters using GridSearchCV or RandomizedSearchCV
11. Document model performance and business interpretation

Recommended Tech Stack

Language	Python 3.x
Libraries	Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
Models	Linear Regression, Random Forest, XGBoost (optional)
Environment	Jupyter Notebook or Google Colab

Deliverables

- Jupyter Notebook covering full ML pipeline
- Model evaluation comparison table
- Feature importance chart
- Actual vs. predicted plot
- Written summary of model performance and limitations

Evaluation Criteria

- Correctness of the ML pipeline steps
- Model selection rationale and comparison depth
- Metric computation accuracy
- Quality of visualizations
- Business interpretation of the model output

Submission Requirements

Platform	Google Drive (assigned folder)
Deadline	End of Week 2
Required Files	Jupyter Notebooks (.ipynb), datasets, PDF exports
Review Session	Present EDA insights and model performance during session

Important Notes

- All source code must be well-commented and follow industry coding standards.
- Submit a README file explaining how to run your project.
- Be prepared to explain your implementation during the review session.
- Original work only — plagiarism will result in disqualification.